1 Real CHIME data



981 bursts · 83 repeaters

CHIME/FRB Catalog 2, 400–800 MHz

→ 950 noise-free templates

→ 856 real noise backgrounds

sources split 60/20/20 by source

2 Pair simulator

lensed



same burst twice

repeat



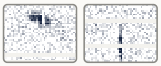
two bursts, one source

impostor



shared envelope / scint.

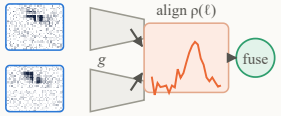
unrelated



two sources

+ nuisances per member: time shift, ΔDM, scattering, scintillation; frozen, hashed sets

3 Siamese correlation net



1. shared encoder, 2-sample time resolution
2. soft time alignment over ±24 samples
3. fuse $|g_A - g_B|$ and $g_A \odot g_B$
4. 4-class head → lensed score s(A, B)

259k parameters

4 Validation, no real positives

- ✓ Real cross-source pairs vs per-pair simulated twins — **FPR ≤ 1.5%**
- ✓ Real-burst copies vs template copies — **ratio 0.97**
- ✓ Second-machine rerun data, sets, training, checks — **7/7 hashes**
- ✓ Grouped folds, sealed test pre-registered protocol — **OOF +0.22**

5 Deployment: where can the detector work?

Timing test on burst lists (false alarms checked on 166,000 scrambles) and survey-geometry simulation → copies are recorded for only ~1.8% of lenses at a transit telescope; an all-sky monitor watching the same sources would find ~50× more lensed repeaters.